streptococcal infection could cross react with myocardial and heart valve tissue, causing cellular destruction.

2. Acute glomerulonephritis

It is often preceded by the skin infection. It is caused by only a few "nephritogenic types (strains)". It develops because some components of glomerular basement membrane are antigenically similar to the cell membranes of nephritogenic streptococci. The antibodies Formed against *Streptococci* cross react with glomerular basement membrane and damage. Some patients develop chronic glomerulonephritis with ultimate kidney failure.

HOTS

Why are some staphylococcal skin infections similar to streptococcal skin infection?

7.4.5 Laboratory Diagnosis

Specimens: Clinical specimens are collected according to the site of lesion. Throat swab, pus or blood is obtained for culture and serum for serology.

Direct Microscopy: Gram stained smears of clinical specimens is done, where Gram positive cocci in chains were observed. It is indicative of streptococcal infection.

Culture: The clinical specimen is inoculated on blood agar medium and incubated at 37° C for 18–24 hours. After incubation period, blood agar medium was observed for zone of beta – haemolysis around colonies.

Catalase test: *Streptococci* are catalase negative which is an important test to differentiate *Streptococci* from *Staphylococci*.

Serology: Serological tests are done for rheumatic fever and glomerulonephritis. It is established by demonstrating high levels of antibody to streptococci toxins. The standard test is antistreptolysin Otitration. ASOtitres higher than 200 units are indicative of prior Streptococcal infection.

7.4.6 Treatment and Prophylaxis

- Penicillin G is the drug of choice.
- In patients allergic to penicillin, erythromycin or cephalexin is used.
- Antibiotics have no effect on established glomerulonephritis and rheumatic fever.
- Prophylaxis is indicated only in the prevention of rheumatic fever, it prevents streptococcal reinjection and further damage to the heart.
- Penicillin is given for a long period in children who have developed early signs of rheumatic fever.



Myth: Eating chocolate encourages the development of acne.

Fact: It is the oils and fats in many chocolate products, and not chocolate itself, that promote sebum production and subsequent acne. Chocolate in low-fat chocolate milk and in fat-free chocolate candies does not encourage acne. Acne suffers do not need to give up chocolate, they need to reduce their lipid consumption.

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